

Energy dissipation analysis of implicit-explicit linear multistep methods for gradient flows



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Abstract

We develop two frameworks for the modified-energy analysis of implicit-explicit linear multistep methods (IMEX-LMMs) for gradient flows: a simple-multiplier framework and its extension to general multipliers. The central idea is to reduce modified-energy dissipation to positivity conditions for associated generating polynomials. With the simple multiplier, this approach reaches order six, and order six is proved to be the highest attainable order. We then go beyond this barrier by introducing general multipliers and a generalized Dahlquist–Baiocchi–Crouzeix equivalence theorem. The resulting framework establishes modified-energy dissipation for IMEX-BDF6 and IMEX-WSBDF7 and yields a new energy-dissipative IMEX-LMM8. To the best of our knowledge, these are the first such results for IMEX-BDF6 and IMEX-LMMs of order higher than six.

01 Gradient flow and IMEX-LMMs

Consider the gradient flow

$$\frac{\partial u}{\partial t} = \mathcal{M}(\mathcal{L}u + f(u)),$$

Here $\mathcal{M}$ and $\mathcal{L}$ are self-adjoint linear operators, negative definite and positive semidefinite, respectively.

The nonnegative potential F satisfies $f(u) = F'(u)$. The energy law is

$$\frac{d}{dt}E[u(t)] \leq 0 \quad \text{with} \quad E[u] = \frac{1}{2}(u, \mathcal{L}u) + (F(u), 1).$$

A k -step IMEX-LMM with step size τ reads

$$\sum_{i=0}^k A_i^{(k)} u^{n+1-i} = \tau \mathcal{M} \left[\sum_{i=0}^k B_i^{(k)} \mathcal{L}u^{n+1-i} + \sum_{i=1}^k \hat{B}_i^{(k)} f(u^{n+1-i}) \right].$$

02 Simple-multiplier framework

Test the IMEX-LMM with the **simple multiplier**

$$\delta u^{n+1} := u^{n+1} - u^n.$$

Let

$$a_i^{(k)} = \sum_{j=0}^i A_j^{(k)}, \quad b_i^{(k)} = \sum_{j=0}^i B_j^{(k)} - 1 + \frac{1}{2}\delta_{i,0}, \quad 0 \leq i \leq k-1,$$

where $\delta_{i,0}$ is the Kronecker delta. Write $\mathbf{a}^{(k)} = [a_0^{(k)}, \dots, a_{k-1}^{(k)}]^\top$ and $\mathbf{b}^{(k)} = [b_0^{(k)}, \dots, b_{k-1}^{(k)}]^\top$.

Define the real-valued generating polynomial associated with $\mathbf{s} = [s_0, \dots, s_{k-1}]^\top$ by

$$T(x; \mathbf{s}) = \sum_{m=0}^{k-1} s_m T_m(x), \quad T_m(\cos \theta) = \cos(m\theta), \quad x \in [-1, 1].$$

Theorem 3.1 in [1]: Under some suitable assumptions on gradient flows, suppose that

$$\min_{x \in [-1, 1]} T(x; \mathbf{a}^{(k)}) \geq \alpha > 0, \quad \min_{x \in [-1, 1]} T(x; \mathbf{b}^{(k)}) \geq \beta > 0.$$

Then, there exist positive semidefinite matrices $G_a, G_b \in \mathbb{R}^{(k-1) \times (k-1)}$ such that the modified energy

$$E_G^n = E[u^n] - \frac{1}{\tau}(\mathbf{v}^n, \mathcal{M}^{-1}\mathbf{v}^n)_{G_a} + (\mathbf{v}^n, \mathcal{L}\mathbf{v}^n)_{G_b} + \ell_f \sum_{i=1}^{k-1} \hat{c}_i^{(k)} \|\delta u^{n+1-i}\|^2$$

satisfies

$$E_G^{n+1} \leq E_G^n, \quad 0 < \tau \leq \tau_{\max}.$$

The step-size bound $\tau_{\max}$ is given in [1, Eq. (3.10)].

Here $\mathbf{v}^n = [\delta u^n, \dots, \delta u^{n+2-k}]^\top$, ℓ_f is the Lipschitz constant of f , $(\mathbf{w}, \mathbf{v})_G = \sum_{i,j} g_{ij}(w^i, v^j)$, and the coefficients $\hat{c}_i^{(k)}$ depend only on $\hat{B}_i^{(k)}$.

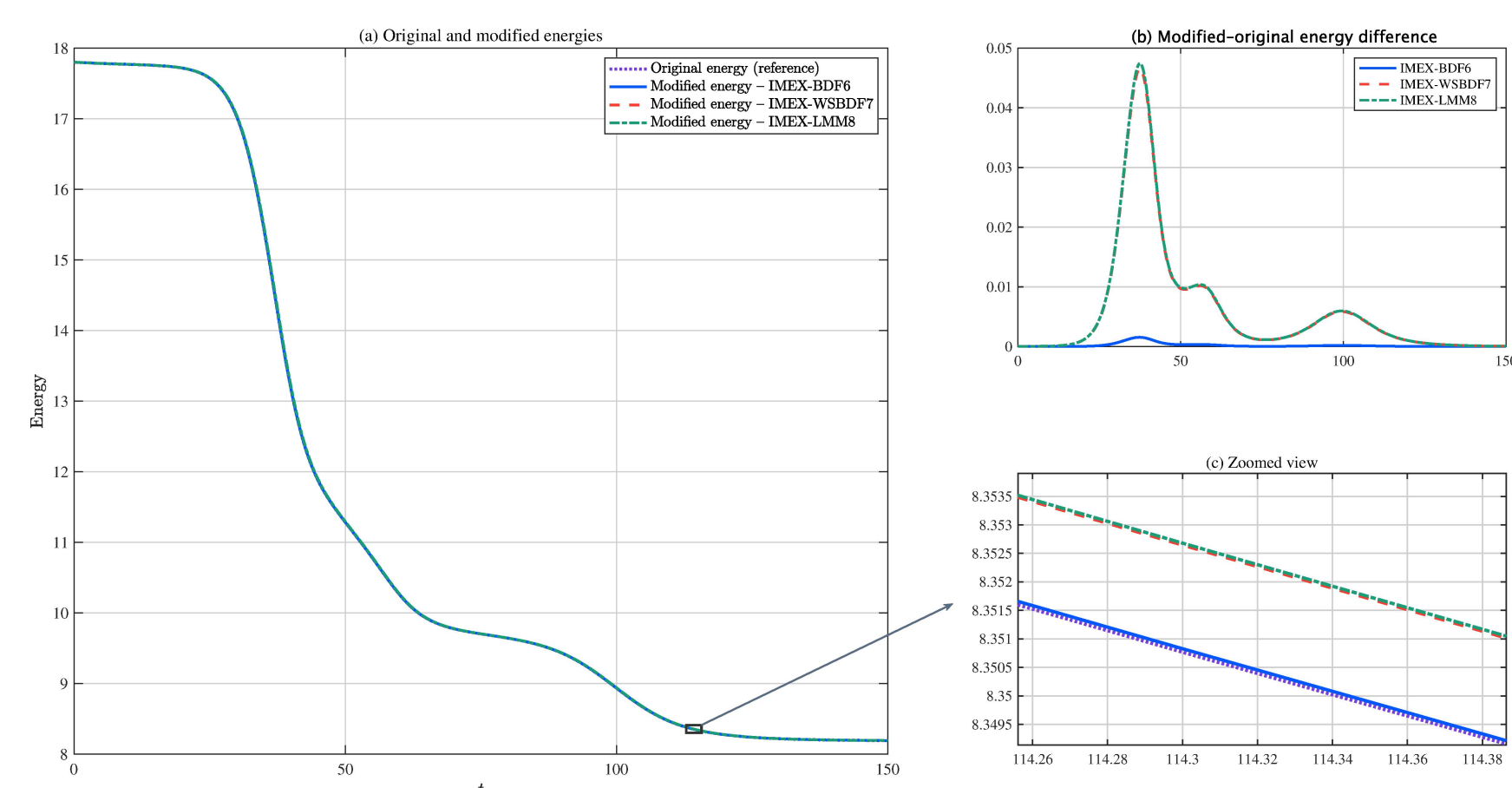
Polynomial conditions
positivity of real-valued
generating polynomials

**Modified-energy
dissipation**
 $E_G^{n+1} \leq E_G^n$

Sixth-order barrier. The simple-multiplier framework reaches order six, while no energy-dissipative IMEX-LMM of order seven or higher exists within this framework.

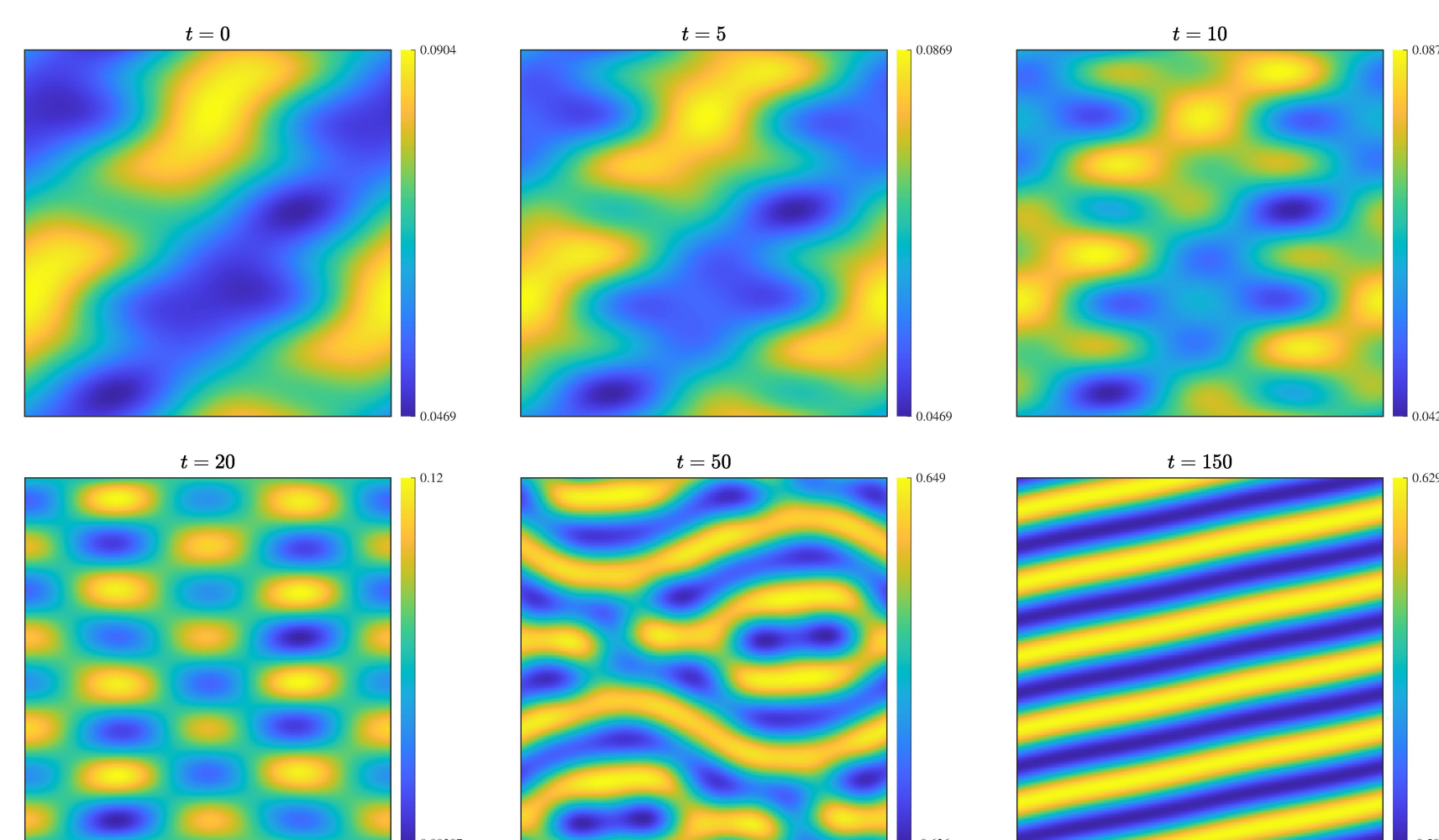
04 Numerical experiments

Energy dissipation



Phase field crystal model with Fourier pseudo-spectral spatial discretization on a 256×256 mesh and time step $\tau = 10^{-3}$. The original energy (reference) is computed by IMEX-LMM8 with time step 10^{-4} .

Pattern evolution



Crystal-growth evolution for the phase-field crystal model, computed by IMEX-LMM8 with $\tau = 10^{-3}$.

03 General-multiplier framework

Test the IMEX-LMM with a **general multiplier**

$$\sum_{i=0}^{k-1} \mu_i \delta u^{n+1-i} \quad \text{with} \quad \sum_{i=0}^{k-1} \mu_i = 1.$$

The generating polynomial associated with $\mathbf{s} = [s_0, \dots, s_r]^\top$ is defined by

$$M(z; \mathbf{s}) = \sum_{i=0}^r s_i z^{r-i}.$$

Write $\boldsymbol{\mu} = [\mu_0, \dots, \mu_{k-1}]^\top$ and $\mathbf{B}^{(k)} = [B_0^{(k)}, \dots, B_k^{(k)}]^\top$.

Theorem 3.7 in [2]: Under some suitable assumptions on gradient flows, assume that $\mu_0 \neq 0$, that $M(z; \mathbf{a}^{(k)})$ and $M(z; \mathbf{B}^{(k)})$ are relatively prime to $M(z; \boldsymbol{\mu})$ and $(z-1)M(z; \boldsymbol{\mu})$, respectively, and that $M(z; \boldsymbol{\mu}) \neq 0$ for $|z| > 1$. Suppose that there exist $\alpha, \beta > 0$ such that for every $\theta \in [0, \pi]$,

$$\operatorname{Re}\{M(e^{i\theta}; \mathbf{a}^{(k)})M(e^{-i\theta}; \boldsymbol{\mu})\} \geq \alpha, \quad \operatorname{Re}\{M(e^{i\theta}; \mathbf{B}^{(k)})(e^{-i\theta} - 1)M(e^{-i\theta}; \boldsymbol{\mu})\} \geq 2\beta(1 - \cos \theta).$$

Then there exist positive definite matrices $G_a \in \mathbb{R}^{(k-1) \times (k-1)}$ and $G_B \in \mathbb{R}^{k \times k}$ such that

$$E_G^n = -\frac{1}{\tau}(\mathbf{v}^n, \mathcal{M}^{-1}\mathbf{v}^n)_{G_a} + (\mathbf{v}^n, \mathcal{L}\mathbf{v}^n)_{G_B} + \sum_{i=0}^{k-1} \mu_i (F(u^{n-i}), 1) + \frac{\ell_f}{2} \sum_{i=1}^{k-1} \hat{c}_i^{(k)} \|\delta u^{n+1-i}\|^2$$

satisfies

$$E_G^{n+1} \leq E_G^n, \quad 0 < \tau \leq \tau_{\max}.$$

Here $\mathbf{u}^n = [u^n, \dots, u^{n+1-k}]^\top$ and the coefficients $\hat{c}_i^{(k)}$ depend only on μ_i and $\hat{B}_i^{(k)}$.

Polynomial conditions
positivity + degree/coprime/root conditions

**Modified-energy
dissipation**
 $E_G^{n+1} \leq E_G^n$

The key ingredient behind Theorem 3.7 in [2] is the following theorem.

Theorem 3.1 in [2]:

Generalized Dahlquist–Baiocchi–Crouzeix equivalence theorem.

Let $\rho(z) = \sum_{i=0}^r \rho_i z^i$ and $\sigma(z) = \sum_{i=0}^r \sigma_i z^i$ be coprime real polynomials of degree $r \geq 1$. For a nonzero real polynomial $\lambda(z) = \sum_{i=0}^r \lambda_i z^i$ of degree at most r and a constant $\gamma \geq 0$, the inequality

$$\operatorname{Re}\{\rho(z)\sigma(\bar{z})\} \geq \gamma|\lambda(z)|^2, \quad |z| \geq 1,$$

is equivalent to the existence of a positive definite matrix $G = (g_{ij}) \in \mathbb{R}^{r \times r}$ and real coefficients $q_i \in \mathbb{R}$ such that, for every sequence in a real Hilbert space,

$$2 \left(\sum_{i=0}^r \rho_i v^{n+i}, \sum_{i=0}^r \sigma_i v^{n+i} \right) - 2\gamma \left\| \sum_{i=0}^r \lambda_i v^{n+i} \right\|^2 = \sum_{i,j=0}^{r-1} g_{ij} (v^{n+1+i}, v^{n+1+j}) - \sum_{i,j=0}^{r-1} g_{ij} (v^{n+i}, v^{n+j}) + \left\| \sum_{i=0}^r q_i v^{n+i} \right\|^2.$$

The introduction of γ and λ is the key extension, and $\gamma = 0$ recovers the classical theorem.

Higher-order methods. Modified-energy dissipation is established for IMEX-BDF6, IMEX-WSBDF7 in [3], and a new eighth-order IMEX-LMM.

Convergence order

τ	IMEX-BDF6		IMEX-WSBDF7		IMEX-LMM8	
	e_{L^2}	rate	e_{L^2}	rate	e_{L^2}	rate
1/20	1.8557e-8	–	8.7777e-9	–	9.6313e-10	–
1/25	4.9686e-9	5.91	1.9096e-9	6.84	1.3546e-10	8.79
1/32	1.1393e-9	5.97	3.4501e-10	6.93	1.8681e-11	8.03
1/40	2.9967e-10	5.98	7.2118e-11	7.01	3.3021e-12	7.77
1/50	7.8649e-11	5.99	1.5045e-11	7.02	5.8202e-13	7.78
1/64	1.7892e-11	6.00	2.6558e-12	7.03	6.9336e-14	8.62

Temporal convergence for the phase field crystal model. Discrete L^2 errors and observed orders are reported for a manufactured solution with exact starting values.

References

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